Lecture 1

Surgical Site Infections, AMR and Why Genomics?

Day 1 · 09:00–09:45 · 45 min
≈ 30 min talk + 15 min discussion

LO1: explain how WGS strengthens SSI and AMR surveillance compared with culture and AST alone

Think of the last surgical patient whose infection didn’t respond to first-line antibiotics.

What did we learn about where it came from?

What is a surgical site infection?

- An infection at or near the incision
- Within **30 days** of surgery (**90 days** with an implant)
- Three levels:
  1. Superficial incisional
  2. Deep incisional
  3. Organ / space

The burden

- The **most common healthcare-associated infection** in low- and middle-income countries (WHO)
- Post-caesarean SSI is a major problem in sub-Saharan Africa, including rural Rwanda
- Consequences:
  - longer hospital stays and reoperation
  - catastrophic costs for households
  - death

SSI after caesarean section in Rwanda

10.9%
Kirehe District Hospital
Nkurunziza et al. 2019

5.7%
Kirehe, later cohort
Velin et al. 2021

6.85%
National pooled estimate
Sibomana et al. 2024

Resistance in SSI isolates from Kirehe

68%
of isolates were Gram-negative

0%
of Gram-negatives were susceptible to ampicillin

92%
were intermediate or resistant to **ceftriaxone**

Velin et al., *Ann Glob Health* 2021. Ceftriaxone is the standard empiric drug.

The gap

- Culture and AST capacity is still limited (Umutesi et al. 2021)
- We **don’t know which resistance genes or clones** are behind these numbers

KEY MESSAGE

That is the gap genomics fills.

DISCUSS

What fraction of SSIs in our hospitals get a culture? An AST?

AMR: the global picture (2019)

1.27 M
deaths **directly caused**
by bacterial AMR

4.95 M
deaths **associated**
with bacterial AMR

- The highest death rate was in **sub-Saharan Africa**
- 5 of the 6 leading pathogens are common causes of SSI

GRAM study, *Lancet* 2022

SSI pathogens to know

Pathogen	Resistance to watch
Staphylococcus aureus	MRSA (*mecA* / *mecC*)
Escherichia coli	ESBLs (CTX-M), carbapenemases
Klebsiella pneumoniae	ESBLs, carbapenemases (KPC, NDM, OXA-48)
Acinetobacter baumannii	Carbapenemases (OXA-23)
Pseudomonas aeruginosa	Efflux, porin loss, VIM / IMP

ESKAPE: *Enterococcus*, *S. aureus*, *Klebsiella*, *Acinetobacter*, *Pseudomonas*, *Enterobacter*

What our current tools tell us

CULTURE AND AST ANSWER

- **Which organism?**
- **Which drugs work?**

Essential for treating **this patient**.

THEY CAN'T ANSWER

- Are these 5 patients infected with the **same strain**?
- Is resistance spreading on a **plasmid** between species?
- Is a known **high-risk clone** circulating?

What a genome adds

A bacterial genome is 2–6 million letters (A, C, G, T). From one genome we get:

1 · Species
confirms or corrects the lab ID

2 · Strain type
MLST: “the bacterium’s surname”

3 · AMR genes
and **where** they sit: chromosome or plasmid

4 · Virulence genes
toxins and other factors

5 · Relatedness
SNP distance: the basis of outbreak detection

Case example: an outbreak response

MRSA on a **neonatal unit** in Cambridge, UK

- **Köser et al., *NEJM* 2012:** separated the linked outbreak isolates from unrelated MRSA that looked identical on routine lab tests
- **Harris et al., *Lancet Infect Dis* 2013:** traced transmission into the community and found a **staff carrier**, which ended the outbreak

KEY MESSAGE

Genomics changed the infection-control decision. We'll use this reasoning in Lab 6.

Genomic surveillance ≠ owning a sequencer



- A reference lab or service provider can do the **sequencing**
- The **expertise stays local**: choosing isolates, recording metadata, analysing, interpreting, communicating
- Africa CDC and national labs are building this network

The next two days

DAY 1 · THE DATA

What sequencing produces, and how to turn it into a genome

DAY 2 · THE ANSWERS

Resistance, strain types, outbreaks, and a sustainable programme

Discussion

15 min · pairs, then the whole room

1. Describe one situation in your work where knowing **“are these infections linked?”** would have changed what you did.
2. What happens to bacterial isolates in our labs after AST is done? **Are they stored?**